Supplementary Materials

Effects of gallium as an additive on activated carbon-supported cobalt catalysts for the synthesis of higher alcohols from syngas

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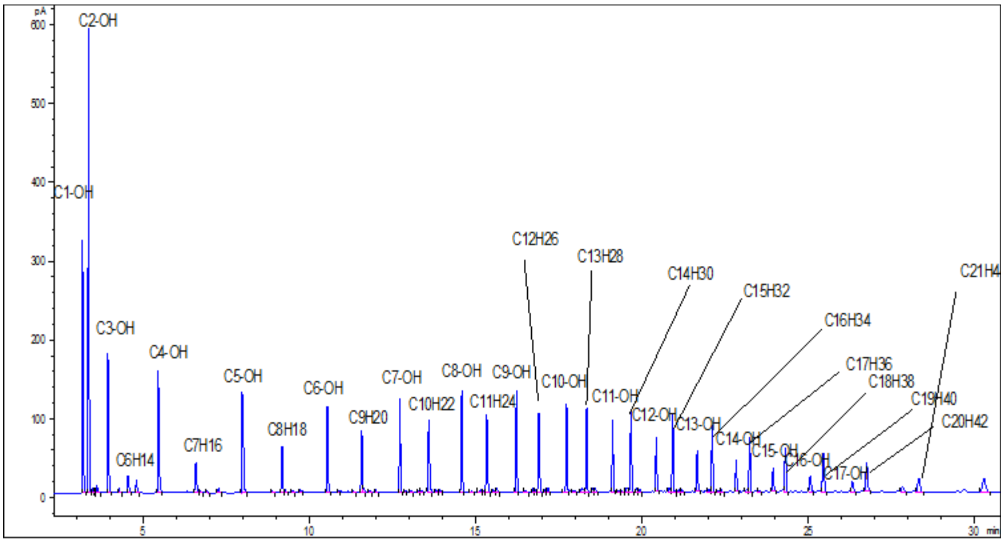

Fig. S1. Off-line GC chart of liquid products collected from cold trap over 15Co–2.0Ga/AC.

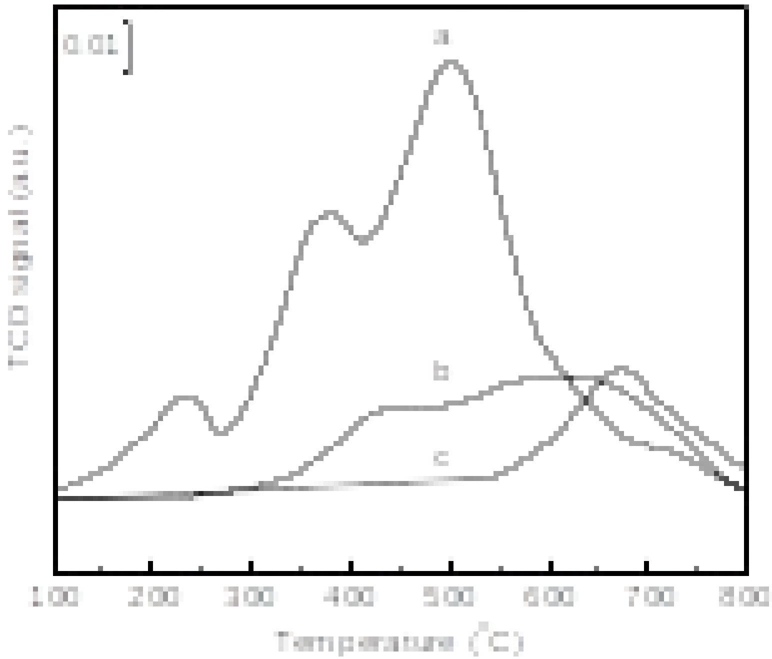

Fig. S2. TPR profiles of a: 15Co–3.0Ga/AC; b: 3.0Ga/AC; c: AC.

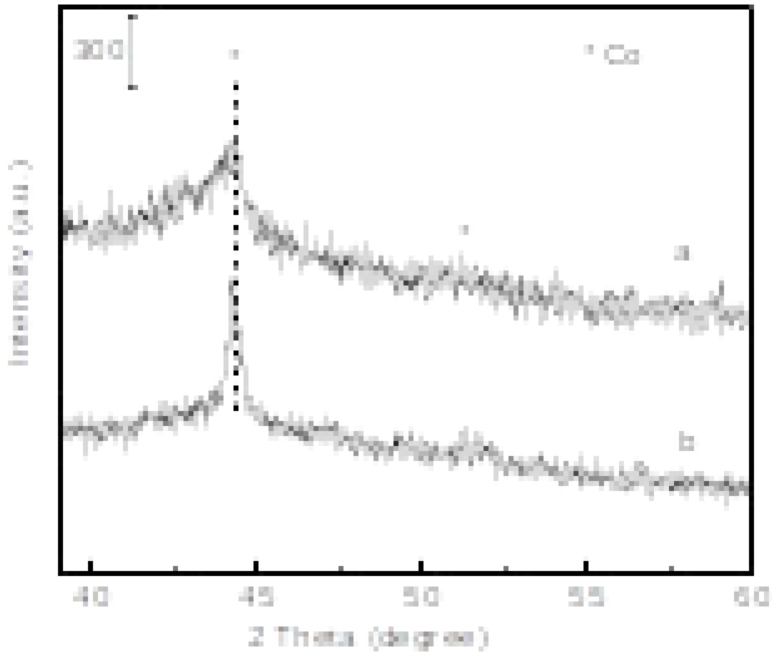

Fig. S3. XRD profiles for the reduced catalysts: a. 15Co/AC and b. 15Co–2.5Ga/AC.

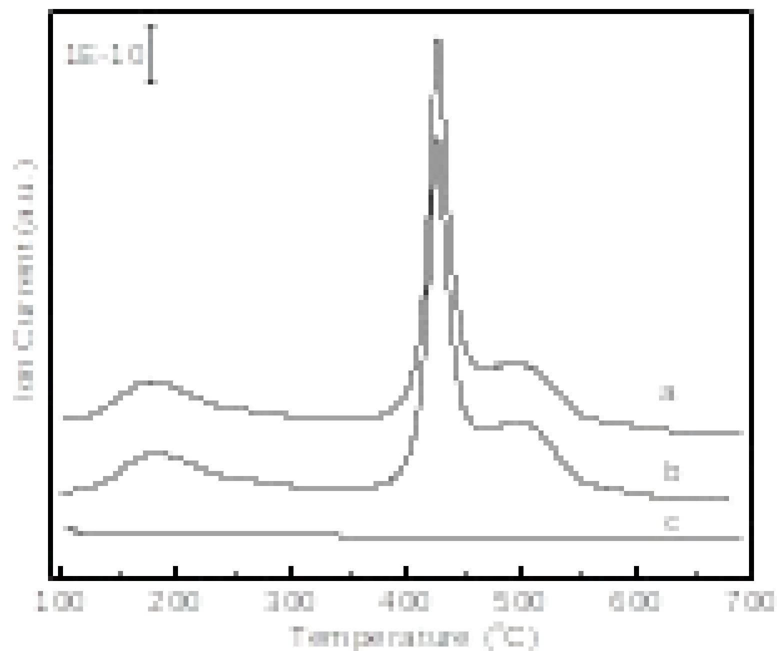

Fig. S4. CO_2-TPD profiles for the fresh catalysts: a. 15Co/AC; b. 15Co–2.5Ga/AC; c. Ga_2O_3.

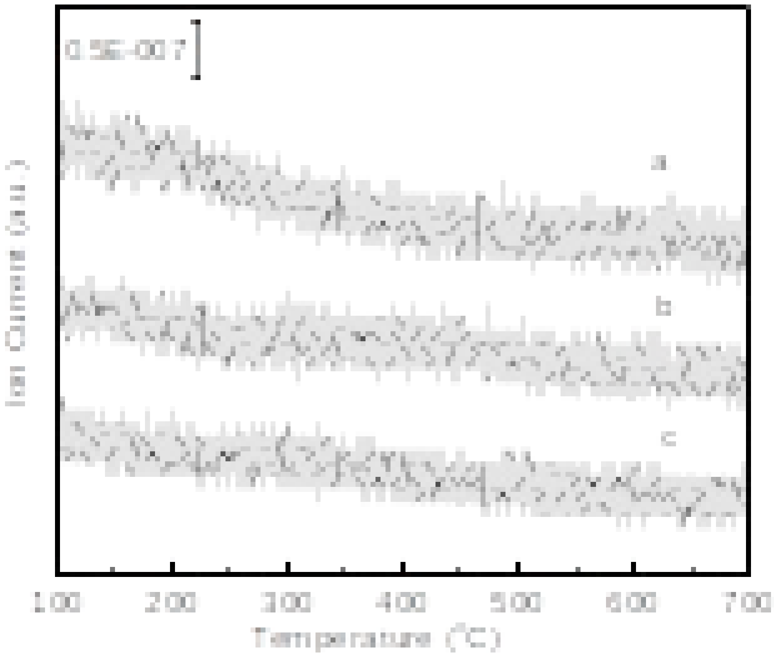

Fig. S5. NH_3-TPD profiles for the fresh catalysts: a. 15Co/AC; b. 15Co–2.5Ga/AC; c. Ga_2O_3.

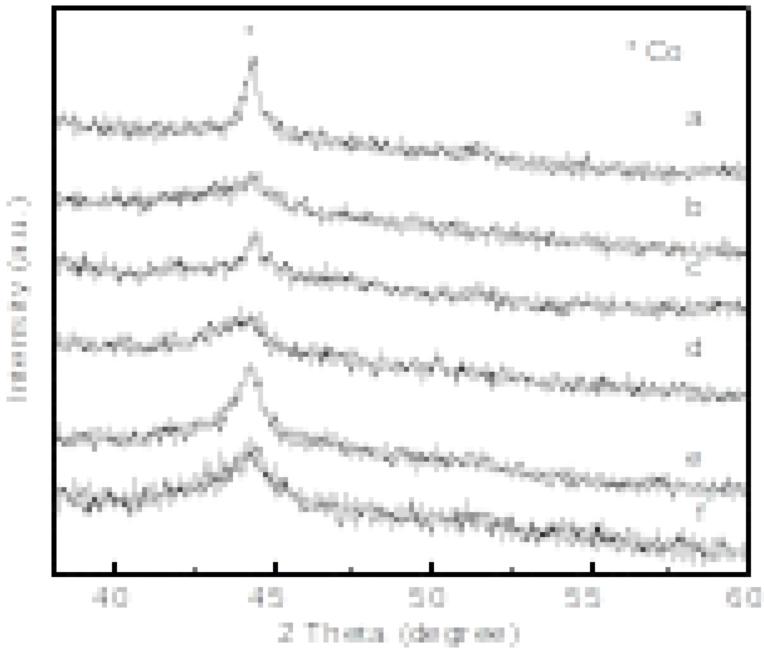

Fig. S6. XRD patterns of the 15Co–*x*Ga/AC catalysts after reaction: (a) 15Co/AC, (b) 15Co–1.5Ga/AC, (c) 15Co–2.0Ga/AC, (d) 15Co–2.5Ga/AC, (e) 15Co–3.0Ga/AC and (f) 15Co–3.5Ga/AC.

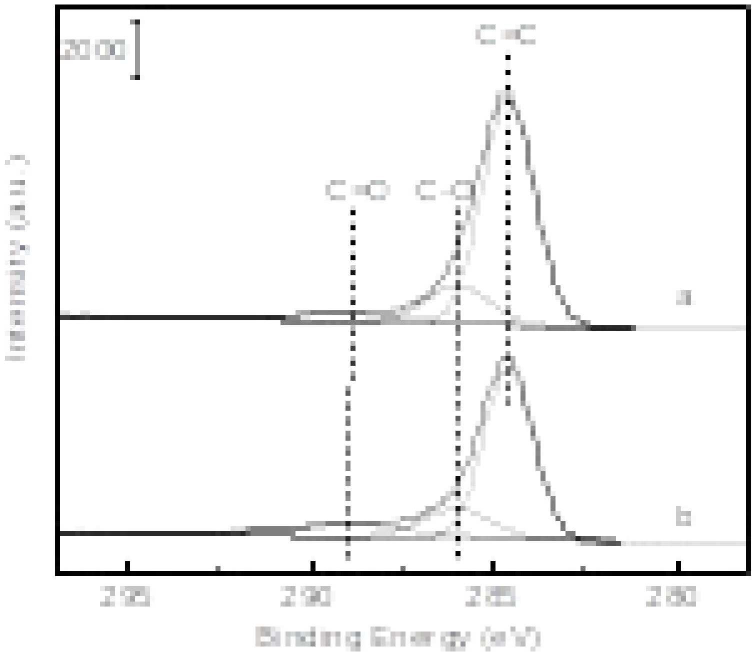

Fig. S7. XPS profiles of C 1s for the spent catalysts: a 15Co/AC and b 15Co–2.5Ga/AC.

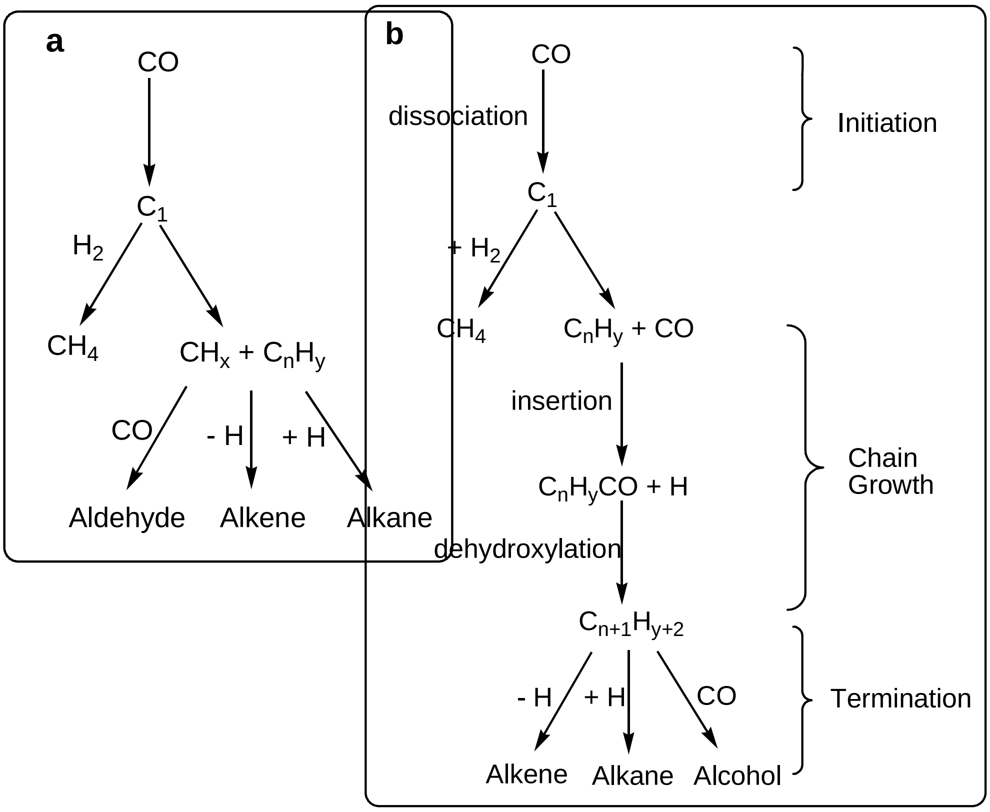

Fig. S8. Plausible reaction mechanism for CO hydrogenation: (a) carbide chain growth mechanism and (b) CO insertion chain growth mechanism.